Know Your Audience: Misreporting Firms' Writing Style When Communicating with Financial Regulators

Eric R. Condie
Assistant Professor
Scheller College of Business
Georgia Institute of Technology
eric.condie@scheller.gatech.edu

September 2022

This paper is based on part of my dissertation at the University of Wisconsin – Madison. I thank my dissertation committee members for their support, guidance, and suggestions: Markus Brauer, Karla Johnstone (Chair), Dan Lynch, Brian Mayhew, and Terry Warfield. I also thank Robbie Moon, Bret Johnson, Zahn Bozanic, and the workshop participants at the University of Wisconsin-Madison, University of Missouri-Columbia, Texas A&M University, Vanderbilt University, Temple University, George Mason University, Portland State University, The Georgia Institute of Technology, and the Brigham Young University Accounting Research Symposium for their helpful suggestions and comments. I think Amanda Convery for her data collection assistance. I gratefully acknowledge funding from the Accounting Doctoral Scholars Program, the Wisconsin School of Business, and the Gil Geis Memorial PhD Scholarship.

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SYNOPSIS: Prior research finds that poor performing and misreporting firms provide annual reports that are less readable and use a different tone than better performing firms to obfuscate the results and avoid a negative perception by financial statement users. Misreporting firms have different incentives when communicating with a regulator that does not yet know that the firm is misreporting. Using SEC comment letters to represent communication between firms and regulators, I examine misreporting firms' comment letter response writing style compared to non-misreporting firms. I find that misreporting firms provide comment letter responses that are more readable than non-misreporting firms, contrary to prior research examining annual reports. I also find that misreporting firms apply a more negative tone in their comment letter responses than non-misreporting firms. Lastly, I find that misreporting firms endure more rounds in the comment letter process. Taken together, these findings indicate that misreporting firms adopt a different writing style in their comment letter responses compared to non-misreporting firms.

Keywords: SEC Comment Letters, Misreporting, Textual Analysis, Readability, Tone

JEL Classifications: M40, M41, M42

Data Availability: All data used are publicly available.

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I. INTRODUCTION

I examine the writing style of misreporting firms compared to non-misreporting firms when responding to SEC comment letters. The management obfuscation literature demonstrates that firms provide less readable disclosures when financial performance is poor (Subramanian et al. 1993; Li 2008) or the financial statements are misreported (Goel, Gangolly, Faerman, and Uzuner 2010; Goel and Gangolly 2012; Lopatta, Jaeschke, and Yi 2014), attempting to dissuade the reader from focusing on negative information. Building upon these findings, I examine the writing style of misreporting firms, whose deceptions have not yet been discovered, when communicating with a financial regulator. Management obfuscation theory (Courtis 2004) predicts that firms use writing techniques (e.g., readability and tone) to obscure the message or confuse the reader when presenting unfavorable information, such as poor financial performance. I examine whether misreporting firms use similar obfuscation techniques when responding to SEC comment letters.

Section 408 of the Sarbanes-Oxley Act of 2002 (SOX) mandates that the Securities and Exchange Commission (SEC) review the filings of publicly-traded firms at least once every three years. If the reviewed filing has omitted required disclosures or if the SEC staff seeks further clarification related to financial reporting issues, the SEC issues a comment letter related to the potential deficiencies. If the SEC staff view the firm's comment letter response as inadequate, additional rounds of comment letters and responses may be necessary before the SEC is satisfied that the matter is resolved.

The SEC comment letter process provides an interesting setting to evaluate the writing techniques used by misreporting firms when communicating with regulators because these firms are unlikely under investigation for this misreporting during the comment letter process. The SEC's Division of Corporate Finance (DCF) reviews public filings, usually within a few months of receipt (Dechow, Lawrence, and Ryans 2016). The SEC's Division of Enforcement (DOE) investigates violations of SEC regulations, such as financial misreporting, and these investigations often occur after substantial time has passed following the misstated periods (Purda and Skillicorn 2015). The SEC likely issues comment letters prior to the DOE commencing an investigation into the misreporting. This setting provides competing incentives for misreporting firms when responding to SEC comment letters. On one hand, misreporting firms may feel compelled to obfuscate their comment letter responses, a similar tactic employed by poor performing firms, to draw investigators' attention away from suspicious information presented in the financial statements. While the DCF review is not investigating the misreporting, these firms know that they are potentially one step away from a DOE investigation.

On the other hand, misreporting firms face incentives to provide clear responses (i.e., avoid obfuscation) to SEC comment letters, desiring to avoid additional scrutiny by the reviewers. This contradicts the incentive for misreporting firms to obfuscate their financial statements because a firm's objective for writing comment letter responses may be different than the objective for writing 10-K narrative disclosures due to differences in the target audience. The firm's target audience of its 10-K is investors and other financial statement users, which are concerned with financial performance. The DCF reviewers are the target audience of comment letter responses, and these reviewers are focused on the adequacy of the financial statement

disclosures.[1] Due to these competing incentives facing misreporting firms when responding to SEC comment letters, the findings in prior literature addressing the use of obfuscation in 10-K writing styles may not hold when examining SEC comment letter responses.

Using misreporting firms identified in SEC Accounting and Auditing Enforcement Releases (AAER) issued between 2005 and 2014, I examine the writing style that misreporting firms use in their comment letter responses compared to non-misreporting firms. I examine comment letter response readability and tone because these two writing styles result from specific word choices by the firms providing the comment letter responses. Misreporting firms could attempt to influence the DCF reviewers through specific word choices that affect the readability and tone of the comment letter responses.

I find that misreporting firms have a different writing style than non-misreporting firms, and that this writing style may lead to a longer comment letter process. I first examine the readability of comment letter responses. I examine readability using the Gunning Fog Index (Gunning 1952), which measures a passage's reading difficulty. The prior literature examining 10-K readability finds that lower readability in annual reports is associated with poor financial performance (Subramanian et al. 1993; Li 2008; Patelli and Pedrini 2014; Patelli and Pedrini 2015) and fraud (Goel et al. 2010; Goel and Gangolly 2012; Lopatta et al. 2014). I compare the comment letter response readability of the sample of misreporting firms to a propensity-score matched sample of non-misreporting firms. After controlling for firm characteristics, firm performance, and comment letter complexity, I find that misreporting firms provide comment

[1] See http://www.sec.gov/divisions/corpfin/cffilingreview.htm.

letter responses that are *more* readable than non-misreporting firms, which is contrary to prior findings examining the readability of annual reports containing negative events.

I next examine the overall comment letter response tone. Firms may use specific tone words to influence perception (Henry 2008) or manage expectations (Allee and DeAngelis 2015). Using the tone keyword list developed by Loughran and McDonald (2011), I separately examine whether the comment letter responses portray each of four different tones: negative, positive, modal strong, and modal weak.[2] Prior literature examining tone finds that disclosures with a higher proportion of positive words are associated with strong financial performance (Henry 2008; Patelli and Pedrini 2014). Conversely, prior literature finds that public filings with a more negative tone are associated with negative events such as internal control material weaknesses (Loughran and McDonald 2011) and illegal acts (Lopatta et al. 2014). Liu and Moffitt (2016) find that firms receiving SEC comment letters using a greater proportion of strong modal language are associated with future restatements. Consistent with the findings of Loughran and McDonald (2011) and Lopatta et al. (2014), I find that misreporting firms provide comment letter responses with a more negative and less persuasive (i.e., strong modal) tone than a matched sample of non-misreporting firms.

My final analysis examines whether misreporting firms' comment letter response writing style influences the comment letter process duration. The previous findings indicate that misreporting firms provide more readable comment letter response and use less persuasive language. They also have a more negative tone, but this may be due to the subject matter of the response if there was a negative event. It's possible that misreporting firms provide response that

[2] Appendix C provides example tone words from the Loughran and McDonald (2011) keyword list for each tone category.

are clearer, but less adequate than non-misreporting firms, leading to more follow-up questions by DCF reviews and more rounds in the comment letter process. Controlling for other factors that affect comment letter process duration, such as the number of filings and number of topics addressed the original comment letter, I find that misreporting firms endure more rounds in the comment letter process than non-misreporting firms, consistent with misreporting firms providing less adequate responses compared to non-misreporting firms.

I also examine the comment letter response writing style for the sample of misreporting firms during the misreporting periods compared to comment letter responses by these same firms during non-misreporting periods. The objective with this analysis is to support my earlier findings that misreporting firms use a different writing style when responding to SEC comment letters during misreporting periods by demonstrating that the sample misreporting firms have a different writing style when responding to comment letters outside of the misreporting period. Consistent with my previous findings, I find that firms provide comment letter responses that are more readable during the misreporting period than outside of the misreporting period. Likewise, the comment letter process is longer for firms during their misreporting period than during their non-misreporting period. Overall, these findings suggest firms adopt a different writing style when responding to comment letters during the misreporting periods compared to non-misreporting periods.

This research contributes to three streams of literature. First, the SEC comment letter literature demonstrates that certain firm characteristics (e.g., prior restatements, high stock volatility, and poor operating results) are significantly associated with the likelihood that a firm will receive a comment letter (Cassell, Dreher, and Myers 2013; Cheng, Gao, Lawrence, and Smith 2014; Ettredge, Johnstone, Stone, and Wang 2011; Kubick, Lynch, Mayberry, and Omer

2016), the likelihood a firm amends or restates its SEC filings (Bozanic, Choudhary, and Merkley 2019; Cassell, Cunningham, and Lisic 2019), and the influence of comment letters on future filings (Bozanic, Dietrich, and Johnson 2017; Johnston and Petacchi 2017; Brown, Tian and Tucker 2018). Second, the textual analysis literature demonstrates that variations in linguistic patterns are associated with management's attempt to portray its message in a particular way to the reader. I build upon these two streams of literature by investigating whether misreporting firms attempt to influence the comment letter process through variations in the readability and tone of their comment letter responses. Lastly, this research contributes to the fraud and misstatement literature. Despite the passage of legislation, such as the Sarbanes-Oxley Act, as well as the issuance of new and revised auditing standards, fraud detection continues to be a problem for auditors, regulators, and the capital markets (Beasley, Carcello, Hermanson, and Neal 2010). This research provides additional insight into the reporting behavior of misreporting firms by examining their comment letter responses.

The remainder of this paper proceeds as follows: Section 2 provides a literature review, Section 3 develops the hypothesis, Section 4 describes the research methodology, Section 5 reports the results, and Section 6 concludes.

II. LITERATURE REVIEW

SEC Comment Letter Process

Cassell et al. (2013) explore the factors that affect the probability of receiving an SEC comment letter, finding that prior restatements, high stock volatility, and poor operating results significantly contribute to the probability of receiving a comment letter. They also find that firms audited by Big 4 or second-tier accounting firms are less likely to receive a comment letter.

Likewise, Moon, Shipman, Swanquist, and Whited (2019) find that firms paying a higher audit fee premium are also less likely to receive a comment letter. Ettredge et al. (2011) find that firms receiving a comment letter due to failure to disclose bad news related to the auditor change are more likely to have weaker corporate governance and less dependence on external financing. Robinson, Xue, and Yu (2011) find that excess CEO compensation is associated with the number of compensation-related disclosure deficiencies identified in comment letters. Lastly, Kubick et al. (2016) find that firms engaging in greater tax avoidance are more likely to receive a tax-related comment letter. Overall, these papers indicate that firms with lower quality governance and financial reporting have a higher probability of receiving a comment letter.

The DCF assigns firm filings to be reviewed by one of 12 DCF offices, each of which focuses on specific industries.[3] The purpose of the Form 10-K and 10-Q reviews is not to audit the reported amounts. The DCF relies on firm management, the auditing profession, and those involved in corporate governance to ensure the financial statement accuracy. Instead, the DCF staff reviews public filings from the perspective of a potential investor, evaluating whether the disclosures appear adequate. If, during the review process, the DCF staff identifies areas where a firm could improve its disclosures, the DCF issues a letter, referred to as a "comment letter", to the firm either requesting a modification of the filing, or requesting the firm provide additional information to the reviewers.

The DCF views the comment letter process as a dialogue between it and the firm (Bozanic et al. 2017). The DCF routinely requests that the firm respond within 10 business days after receiving a comment letter. If the response is not adequate to address the issues raised by

[3] See https://www.sec.gov/info/edgar/siccodes.htm for the SIC codes assigned to each DCF office.

the reviewers, additional rounds of comment letters and responses occur until the DCF staff are satisfied with the firm's response. Bozonic et al. (2019) find that firms engaging external legal counsel experience more rounds of SEC comments before final resolution. Likewise, Cassell et al. (2019) find that the readability of comment letter responses is negatively associated with the duration of the comment letter process, measured in number of days and number of rounds.

In contrast to the Division of Corporate Finance, whose focus is to review disclosures in public filings for compliance with securities regulations, the purpose of the Division of Enforcement (DOE) is to investigate and prosecute violations of securities laws. Information leading to DOE investigations comes from a variety of sources, including whistleblowers (SEC 2015, page 8), law enforcement, the press, and the results of periodic inspections performed by the DCF (Feroz, Park, and Pastena 1991). Bozanic et al. (2017) note that if a firm and the DFC cannot come to agreement regarding certain disclosures during the review process, the DFC refers the matter to the DOE for investigation. In many cases, the DOE publicly announces the commencement of an investigation, as well as the result of such investigation through Accounting and Auditing Enforcement Releases (AAER), Litigation Releases, and notices of Administrative Proceedings.[4]

The DFC reviews selected filings promptly after they are received, whereas the DOE opens an investigation when relevant information becomes available. A significant timing difference potentially exists between the activities of these two divisions related to the same reporting period. Dechow et al. (2016) find that the median number of days between the fiscal year-end and the first comment letter is 193, or approximately 6.4 months. Conversely, Purda

[4] See https://www.sec.gov/litigation.shtml

and Skillicorn (2015) find that the average length of time between the end of the fraud period and the AAER issuance is approximately 3.4 years. Figure 1 provides an example timeline of comment letter and AAER issuances.

[INSERT FIGURE 1 HERE]

Writing Styles in Financial Reporting

Management has an incentive to obfuscate certain disclosures when it does not want to reveal the true nature of an issue to the public for fear of negative publicity or market reaction (Courtis 2004). In general, "obfuscation" is defined as ''a narrative writing technique that obscures the intended message, or confuses, distracts or perplexes readers, leaving them bewildered or muddled'' (Courtis 2004). The management obfuscation hypothesis applied to financial reporting is a form of impression management where a firm emphasizes successes and obfuscates failures (Patelli and Pedrini 2014).

Often, the management obfuscation literature measures the readability of disclosures as a proxy for management's intent to not provide a clear explanation of an issue (Merkl-Davies and Brennan 2007). The readability literature shows that lower readability in annual reports is associated with poor financial performance (Subramanian et al. 1993; Li 2008; Patelli and Pedrini 2014; Patelli and Pedrini 2015) and future financial difficulty (Smith and Taffler 1992). Laksmana, Tietz, and Yang (2012) examine the compensation discussion and analysis (CDA) section of the proxy statement and find that firms with overcompensated CEOs also provide less readable CDAs. The authors note that providing less readable compensation disclosures may be a way to manage negative impressions related to overcompensation of executives. Prior literature also finds that financial statements with lower readability are associated with fraud (Goel et al. 2010; Goel and Gangolly 2012) and illegal acts (Lopatta et al. 2014).

Firms may also attempt to influence readers' perceptions of their public filings by altering the tone of a particular narrative or the entire filing. Patelli and Pedrini (2014) find that an optimistic tone (i.e., more use of words that praise an individual or entity than words that blame) in the letter to shareholders is associated with strong future performance. Patelli and Pedrini (2015) further find that aggressive financial reporting is associated with letters to shareholders that use a higher proportion of words indicating certainty and more use of complex words.[5]

Henry (2008) finds that firms that have a greater proportion of "positive" words compared to "negative" words in their earnings press releases are associated with positive abnormal stock returns, even after controlling for firm performance. This finding suggests that narrative tone influences the reader's perception of the information. Lopatta et al. (2014) find that firms violating the Foreign Corrupt Practices Act use more words categorized as "negative", "litigious", and "uncertain" in their 10-K filings compared to non-violating firms. Loughran and McDonald (2011) find that negative and strong modal tone words used in 10-K filings are associated with future internal control weaknesses. Purda and Skillicorn (2015) find that fraud firms use a higher proportion of "uncertain" words compared to non-fraud firms in the management discussion and analysis section of the 10-K.

III. HYPOTHESIS DEVELOPMENT

Managers' responses to SEC comment letters provide a relevant setting to examine firms' writing style when communicating with a regulator. Previous research examining obfuscation

[5] See Appendix C for examples of words classified into the following "tones": positive, negative, strong modal, and weak modal.

theory focuses on firms' impression management regarding the information presented in the 10-K (Merkl-Davies and Brennan 2007; Li 2008; Goel and Gangolly 2012; Patelli and Pedrini 2014). In these public filings, the financial statement numbers communicate the firm's financial performance. Management attempts, through variation in readability or tone of the narrative portion of the annual report and other communications, to emphasize successes and obfuscate information describing unfavorable results. The firm's target audience of its 10-K is investors and other financial statement users.

Misreporting firms' writing style when in their SEC comment letter responses may differ from the writing style used in the 10-K for several reasons. First, contrary to prior management obfuscation research, the target audience of comment letter responses is the SEC's Division of Corporate Finance (DCF). Although the comment letter responses will eventually be made public, the incentive to maximize stock performance and attract potential investors or other sources of capital is less when responding to SEC comment letters than it is when reporting financial results. Second, the scope of a DCF review is less than that of a financial statement audit, potentially decreasing the incentive for misstating firms to hide information from the reviewers.[6] Third, if a firm is intentionally misstating its financial statements, the average time between the misstatement event and an initial AAER is over three years (Dechow et al. 2016). In contrast, the average time between year-end and the issuance of a comment letter is just over six months (Purda and Skillicorn 2015), meaning that the DCF will begin reviewing the 10-K or 10-Q filing shortly after the filing is made (i.e., two to three months after period end). Therefore, it is unlikely that the DCF review and DOE investigation are occurring simultaneously, with the

[6] See http://www.sec.gov/divisions/corpfin/cffilingreview.htm

DOE investigation probably occurring well after the DCF review is complete. Thus, a firm is likely less concerned with obfuscation when responding to SEC comment letters than it is when issuing financial statements. Misreporting firms may want to provide more readable language in comment letter responses to avoid additional scrutiny due to unclear responses.

Conversely, a firm faces several incentives to obfuscate the information communicated to the DCF through its comment letter responses. First, the DCF reviews are not the same as DOE investigations. However, firm management knows that when one branch of the SEC (i.e., DCF) is closely reviewing the filing, they are potentially only one step away from a DOE investigation. Second, although not the primary audience to comment letter responses, the public eventually has access to these responses shortly after the comment letter process ends. An AAER firm is incentivized to provide responses that obscure clues that it is engaging in inappropriate conduct. Overall, AAER firms have an incentive to employ impression management techniques, including obfuscation, in comment letter responses, attempting to defuse the discovery risk and to draw attention away from misstated financial information.

Similar claims can be made regarding the misreporting firms' comment letter response tone. The findings related to 10-K tone shown in previous fraud research may not hold when examining comment letter response because misreporting firms' incentives to present the information a certain way when responding to comment letters are different than when presenting information for external reporting purposes. Given these arguments, I propose the following non-directional hypothesis:

H1: Misreporting firms' comment letter response writing style differs from that of non-misreporting firms.

IV. METHODOLOGY

Sample Selection

Misreporting Firms

To examine whether misreporting firms employ obfuscation techniques when responding to comment letters, I start by identifying all firms that were the subject of an SEC Accounting and Auditing Enforcement Release (AAER) issued between January 2005 and September 11, 2014 (AAERs 2358 – 3582), which is the cut-off for the AAER data-collection.[7] I begin the AAER data collection in 2005 as the SEC did not start making the comment letter conversations publicly available until August 2004.[8] For these AAERs, I read each of the associated AAERs[9,10] and hand-collect the firm name, the misstated SEC filings, and the SEC securities regulation violations indicated.

From the AAERs examined, I identify 339 firms that were implicated in an AAER between January 2005 and September 11, 2014. From this population of AAER firms, I eliminate 10 firms lacking CIK numbers, four firms because the AAER only implicates the auditor, and six firms because the AAER is not related to a misstated 10-K or 10-Q. After

[7] The SEC issues AAERs to firms implicated for committing financial reporting fraud (i.e., intentional misreporting) for violating Section 17(a) of the Securities Act of 1933 or Rule 10b-5 of the Securities Exchange Act of 1934, which are considered the SEC's anti-fraud provisions of their respective regulations. The SEC also issues AAERs to firms that are implicated for financial misreporting, but the SEC does not identify them as violating Section 17(a) or Rule 10b-5. Dechow, Ge, Larson, and Sloan. (2011) state that fraud is often implied in the issuance of an AAER, as the misreporting is of sufficient severity for investigation and public notification through AAER issuance. Therefore, I include all firms in the sample that are implicated in AAERs issued during the sample period.
[8] See SEC Press Release 2004-89 at https://www.sec.gov/news/press/2004-89.htm
[9] Often, the SEC Division of Enforcement will issue multiple AAERs related to a single misstatement event.
[10] For the AAER identification process, I relied on the AAER dataset compiled by the University of California – Berkeley Center for Financial Reporting and Management. This dataset contains all AAERs issued between May 17, 1982 and August 31, 2012. The AAERs in the dataset are grouped by firm and misstatement event, identifying the misstated 10-K and 10-Q filings. This dataset was created for use in Dechow et al. (2011), where it is described in more detail.

removing these firms, the sample includes 319 misreporting firms where the AAER is related to a misstated 10-K or 10-Q. Table 1 contains the summary of the sample selection procedures.

Comment Letter Conversations

I obtain the SEC comment letter information from the Audit Analytics (AA) comment letter and comment letter conversations databases. Upon merging the misreporting firm list with the AA comment letter database, I eliminate 66 misreporting firms that did not receive any comment letters during the sample period. I identify the comment letters relating to the 10-K or 10-Q filing that is the subject of an AAER and eliminate 164 misreporting firms that did not receive any comment letters related to a misstated 10-K or 10-Q. I also eliminate nine AAER-related comment letter conversations without sufficient data available to calculate the control variables. Finally, 23 AAER-related comment letter conversations related to firms receiving multiple comment letters applying to different filings during the misreporting period. The final sample size of AAER-related comment letter conversations is 103.

[INSERT TABLE 1 HERE]

Control Sample

I select the control sample to test the hypotheses from the population of comment letter conversations available through Audit Analytics. The AA comment letter conversations database contains 95,926 comment letter conversations as of June 9, 2016. Many of these comment letter conversations address multiple filings made by the subject firm. To simplify the process of obtaining relevant control variables in the statistical models and to enhance comparability among observations, I focus on comment letter conversations where at least one 10-K filing is the subject of the comment letter. This methodology is consistent with existing comment letter research (Cassell et al. 2013; Cassell et al. 2019, Bozanic et al. 2019; Bozanic et al. 2017;

Dechow et al. 2016; Brown et al. 2018).[11] I eliminate 66,555 conversations that do not relate to a 10-K filing. Following Cassell et al. (2019), I eliminate 12,689 comment letter conversations that appear to be incomplete in Audit Analytics.[12] I also eliminate 2,580 comment letter conversations from the control sample that do not contain sufficient data to calculate the control variables. Lastly, I eliminate the AAER-related comment letter conversations (N = 103) previously identified from the control sample population. The remaining sample size of non-AAER-related comment letter conversations is 13,999. As noted in Table 2, the misreporting firm sample includes comment letter responses related to misstated filing years between 2003 and 2011. For the matched-sample analysis, I further eliminate 3,687 comment letter conversations not related to these years. The final sample size of non-AAER-related comment letter conversations subject to the propensity-score matching procedures is 10,312.

Research Design

The hypotheses concern the writing style (i.e., readability and tone) of a firm's response to SEC comment letter inquiries. Often, if the comment letter process consists of several rounds, subsequent comment letters and responses focus on specific points not adequately addressed in previous responses. I examine the firm's initial response to an SEC comment letter as it usually provides the greatest amount of information. Therefore, I collect the first firm response in the comment letter conversation that provides substantial information to perform the textual analysis.[13]

[11] For AAER firms where the AAER solely relates to a 10-Q filing (N = 5), I select the control variable information from the 10-K pertaining to the same fiscal year.
[12] See Appendix D for procedures to identify incomplete comment letter conversations.
[13] See Appendix D for procedures to identify first substantial comment letter response.

The hypothesis predicts a difference in the comment letter response writing style between misreporting firms and non-misreporting firms. I use the following model to examine this hypothesis:

$$\begin{aligned} CL_WRITING_STYLE_{it} = \beta_0 &+ \beta_1 AAER_{it} + \beta_2 SIZE_{it} + \beta_3 AGE_{it} + \beta_4 NUM_FILINGS_{it} \\ &+ \beta_5 NUM_TOPICS_{it} + \beta_6 SEC_WORD_COUNT_{it} + \beta_7 RESTATE_{it} \\ &+ \beta_8 LOSS_{it} + \beta_9 BANKRUPTCY_RANK_{it} + \beta_{10} EXTFINANCING_{it} \\ &+ \beta_{11} BMV_{it} + \beta_{12} ROA_{it} + \beta_{13} BIG4_{it} + \beta_{14} MULTIPLE_CL_{it} \\ &+ \beta_j IndustryFE + \beta_k YearFE + \varepsilon_{it} \end{aligned} \quad [1]$$

I use two different proxies to examine the dependent variable, *CL_WRITING_STYLE*: *READABILITY* and *TONE*, as described below.

Comment Letter Response Readability

READABILITY is measured using the Fog index, which is common in the textual analysis literature (Cassell et al. 2019; Goel et al. 2010; Burgoon, Mayew, Giboney, Elkins, Moffitt, Dorn, Byrd, and Spitzley 2015; Lopatta et al. 2014; Li 2008; Laksmana et al. 2012). The Fog index was created by Robert Gunning (Gunning 1952) and is a function of the average number of words per sentence and the percentage of difficult words in the passage.[14] Higher Fog index scores indicate less readable text. I calculate the Fog index using the Python Textstat package applied to the CL_TEXT field in the AA comment letter database.[15] For ease of interpretation and consistent with Cassell et al. (2019), I multiply the Fog index by negative one so that larger *READABILITY* values indicate more readable text. Therefore, positive regression coefficient values indicate more readable responses.

[14] The fog index is calculated as Fog = (Average Words per Sentence + Percentage of Difficult Words) * 0.4. Difficult words are those with three or more syllables.

[15] Appendix B provides examples of comment letter responses of differing readability levels, calculated using the Fog index.

The main independent variable in model [1] is *AAER*, which is equal to one if the comment letter response relates to a misstated 10-K or 10-Q filing, or zero otherwise. If misreporting firms provide less readable comment letter responses, this supports the argument that misreporting firms implement obfuscation techniques attempting to confuse the reader. Conversely, if misreporting firms provide more readable comment letter responses, this supports the argument that misreporting firms provide comparatively more clear responses to avoid additional questions due to obscure language in the comment letter response. I do not make a directional prediction for *AAER*.

Certain other factors related to a firm's characteristics, its performance, or the complexity of the comment letter process will likely contribute to a firm's comment letter response readability, which I control for in model [1]. Firm-related characteristics affect the complexity of issues included in a firm's financial statements, and these complex issues may be the subject of an SEC comment letter. Following Li (2008) and Cassell et al. (2019), I control for firm size (*SIZE*), firm age (*AGE*) and the need for external financing (*EXTFINANCING*).

Performance-related issues could also affect the readability of comment letter responses. The textual analysis literature examining financial statement readability (Subramanian et al. 1993; Clatworthy and Jones 2001; Smith and Taffler 1992) find that poor performing firms produce less readable financial statements. The incentive to decrease readability due to poor financial performance could also affect comment letter response readability. I control for whether the firm reported a loss (*LOSS*) in the current year or announced a restatement (*RESTATE*) within the previous two years. I also control for other performance-related factors such as the firm's book-to-market value (*BMV*) and return on assets (*ROA*), as well as the firm's

overall financial health (*BANKRUPTCY_RANK*), which is the decile ranking of a firm's Altman Z-score.

The comment letter process itself could affect the firm's comment letter response readability. I proxy for comment letter complexity through the inclusion of control variables for the number of filings (*NUM_FILINGS*), number of topics (*NUM_TOPICS*), and number of words (*SEC_WORD_COUNT*) included in the initial SEC comment letter. A firm's auditor could potentially influence the writing style of a firm's comment letter response. I include an indicator variable equal to one if the firm is audited by a Big 4 (*BIG4)* auditor. As noted in Table 1, several firms received more than one comment letter related to different filings during the misreported period, which may influence the regression results. I include a control variable *MULTIPLE_CL*, which is equal to one if the firm is represented more than once in the sample, and zero otherwise to control for potential data independence issues. Lastly, I control for the influence of fiscal year (*YearFE*) and the specific DCF reviewing office, which is determined by the firm's industry (*IndustryFE*). Appendix A contains all the variable definitions.

Comment Letter Response Tone

The comment letter response tone (*TONE*) reflects the type of words a firm chooses to include in its comment letter response. I measure *TONE* using the keyword list created in Loughran and McDonald (2011)[16], which includes several word dictionaries, adapted for a financial reporting setting. Specifically, I separately examine the use of negative, positive, strong modal and weak modal tone words. Negative and positive tone words are commonly used in the tone literature (e.g., Henry 2008, Lopatta et. al. 2014). Strong modal words express obligation,

[16] The keyword list is available on Bill McDonald's website at http://www3.nd.edu/~mcdonald/Word_Lists.html.

certainty, or intensity, whereas weak modal words express possibility (Loughran and McDonald 2011). Strong and weak modal words are persuasive in nature and could be used to attempt to influence the reader toward a certain conclusion.[17] Consistent with Lopatta et al. (2014), I calculate *TONE* as the number of specific tone words divided by the total number of words in the firm's first substantial comment letter response. The percentage of specific tone words to total words indicates the strength of a particular tone in the subject text. All other variables are previously defined.

Comment letter response tone has not been extensively examined in the comment letter literature and I do not make a signed prediction for the *AAER* coefficient associated for each tone category. The literature examining the tone of fraudulent financial statements finds that these contain comparatively higher proportions of words classified as "negative", "litigious", and "uncertain" (Lopatta et al. 2014; Purda and Skillicorn 2015), but the comment letter response writing style may not reflect these results.

V. RESULTS

Descriptive Statistics

In Table 2, I report descriptive statistics. Panel A reports the year of the 10-K that is targeted in each SEC comment letter in the sample, grouped by AAER and non-AAER firm observations. Panel B reports the industry representation of the firms in the sample, separated by AAER and non-AAER firms. When considering all observations, the top three industries represented are durable manufacturers (18 percent), computers (14 percent), and financial

[17] Appendix C provides example tone words from the Loughran and McDonald (2011) keyword list for each tone category.

institutions (13 percent). These industries also represent the top three industries represented in each of the AAER and non-AAER samples.

Table 2, Panel C reports the mean differences between the AAER and non-AAER observations prior to selecting a matched sample. I report *AGE* and *SEC_WORD_COUNT* in their unlogged form for ease of interpretation. Panel C also includes *ROUNDS*, which is the total number of rounds in the comment letter process, comprising the initial SEC comment letter and follow-up inquiries during the comment letter process. Panel C reports that, for variables reported at their means, misreporting firms (*AAER*=1) receive SEC comment letters that are longer (*SEC_WORD_COUNT*; $p<0.01$), address more filings (*NUM_FILINGS*; $p<0.01$), and address more topics (*NUM_TOPICS*; $p<0.01$) than non-misreporting firms. Further, misreporting firms endure more rounds (*ROUNDS*) in the comment letter process $p<0.01$) compared to non-misreporting firms. These differences suggest that misreporting firms' public filings comparatively have more issues that the DCF target for review, and misreporting firms endure a more arduous comment letter process compared to non-misreporting firms.

I also find that the mean *READABILITY* of misreporting firm comment letter responses is higher than the mean *READABILITY* for non-misreporting firms ($p<0.05$). Misreporting firms also comparatively provide comment letter responses with a higher mean proportion of negative words (*NEGATIVE*) than non-misreporting firms ($p<0.01$).

[INSERT TABLE 2 HERE]

Table 3 reports the correlation coefficients among the independent variables. The variable of interest (*AAER*) is positively and significantly ($p<0.05$) correlated with *READABILITY* and *NEGATIVE*. Each of the SEC comment letter complexity variables (*NUM_FILINGS*, *NUM_TOPICS*, *SEC_WORD_COUNT*) are significantly ($p<0.05$) correlated with

READABILITY, which highlights the importance of controlling for comment letter complexity in the regression models. In untabulated results, the mean and maximum variance inflation factor (VIF) are 1.71 and 3.93, respectively, indicating that the statistical analysis is not significantly affected by multicollinearity.

[INSERT TABLE 3 HERE]

Results of Hypothesis Testing

In the sample population, AAER firm comment letter responses represent approximately one percent of the total comment letter observations. As the AAER observations are such small percentage of the total sample population, I select a propensity-score matched sample from the population of non-AAER firms based on the following logit fraud prediction model adapted from Lennox and Pittman (2010):

$$P(AAER_{it} = 1) = \beta_0 + \beta_1 SIZE_{it} + \beta_2 AGE_{it} + \beta_3 LOSS_{it} + \beta_4 BANKRUPTCY_RANK_{it} + \beta_5 BIG4_{it} + \beta_6 AUDITOR_TENURE_{it} + \beta_7 M\&A_{it} + \beta_8 NEG_BOOK_EQ_{it} + \beta_9 DEBT_EQ_ISSUE_{it} + \beta_j IndustryFE + \beta_k YearFE + \varepsilon_{it} \quad [2]$$

The independent variables *SIZE*, *AGE*, *LOSS*, *BANKRUPTCY_RANK*, and *BIG4* are previously defined. The model also controls for the length that a firm has been audited by the same auditor (*AUDITOR_TENURE*), whether the firm had a merger or acquisition in the current year (*M&A*), whether the firm has more liabilities than assets (*NEG_BOOK_EQ*), and whether the firm issued debt or equity (*DEBT_EQ_ISSUE*) in the current year. Table 4, Panel A reports the results of this first-stage regression. Propensity-score matched sample observations are selected using a caliper of 0.03 and without replacement. Panel A reports that the area under the ROC curve is 0.841. The resulting sample includes 103 misreporting firm comment letter responses with a propensity-score matched group of 103 non-misreporting firm comment letter

responses, for a total sample size of 206.[18] Panel B reports the mean differences of the propensity-score matched sample for the covariates from Model [5]. The mean differences between the AAER sample and non-AAER sample for each variable in the model are not statistically significant, which suggests I have good covariate balance.

[INSERT TABLE 4 HERE]

Comment Letter Response Readability

Table 5 reports the results of the analysis of the association between *READABILITY* and *AAER* using the propensity-score matched sample. H1 predicts that the writing style, including readability, of a firm's comment letter response will be different for misreporting firms compared to non-misreporting firms. The fraud detection literature analyzing fraudulent 10-K readability finds that fraud firms submit 10-Ks that are less readable than non-fraud firms (Goel et al. 2010; Goel and Gangolly 2012; Lopatta et al. 2014). Interestingly, I find that *AAER* is positively associated with *READABILITY* ($p<0.05$), suggesting that misreporting firms provide comment letter responses that are *more* readable than non-misreporting firms, which is contrary to these prior findings. As reported in Table 2, Panel C, misreporting firms endure more rounds in the comment letter process than non-misreporting firms. The findings that misreporting firms provide more readable comment letter responses than non-misreporting firms is consistent with

[18] There is a risk that the control sample of non-misreporting firms contains firms that are also misreporting their financial statements, but have not yet been caught. I searched through all AAERs issued subsequent to the sample period through mid-2017 to verify that none of the control firms were implicated for financial misreporting during the sample period. I further performed an Internet search for the control firms to determine whether they misreported their financial statements during the sample period, but were not cited in an AAER. As 2011 is the final year represented in the sample (see Table 2, Panel A), the probability is high that significant misreporting will have come to light by 2017 (the period the subsequent AAER search was performed). Significant misreporting schemes often require coordination among several individuals, which makes it difficult to perpetuate a misreporting scheme indefinitely. None of the control firms were implicated for financial statement misreporting during the sample period. Two firms were implicated for violating the Foreign Corrupt Practices Act several years subsequent to the sample period. The results are unchanged with the exclusion of these control firms.

misreporting firms attempting to avoid additional attention by providing comment letter responses that are comparatively clearer, but possibly less adequate, leading to additional rounds of inquiry in the comment letter process. [19]

Supplemental Readability Analysis: In addition to the main analysis in Table 5 examining the association between misreporting firms and comment letter response readability, I also examine other factors that may influence comment letter response readability. First, I examine the association between misreporting firms and comment letter response readability, controlling for SEC comment letter readability. There are two main reasons why it is important to consider the association between SEC comment letter readability and response readability in the regression model. First, the initial SEC comment letter readability may influence the firm's response readability. Second, firms will often, but not always, include the question from the SEC comment letter as part of their comment letter response. Therefore, I control for SEC comment letter readability in examining model [1]. I find (untabulated) a statistically significant and positive association ($p<0.01$) between SEC comment letter readability and *READABILITY*, consistent with the readability of the initial comment letter influencing the readability of the comment letter response. After controlling for SEC comment letter readability, I still find a positive association ($p=0.05$) between *AAER* and *READABILITY*.

I also examine whether CEO or CFO involvement in the misstatement or the existence of proprietary information – determined by firms requesting confidential treatment of their comment letter responses and firms with a high proportion of research and development expense – influence the comment letter readability. I obtain CEO and CFO misstatement involvement by

[19] The results in Table 5 show that the control variables used in the propensity-score matching regression (*SIZE, AGE, LOSS, BANKRUPTCY_RANK,* and *BIG4*) are not significantly associated with *READABILITY,* indicating that the sample is sufficiently matched on these variables.

reading each AAER and noting whether the CEO or CFO involvement was cited in the AAER. For the misstating firms, the CEO or CFO was involved in the misstatement 74 percent of the time, consistent with the findings in Beasley et. al. (2010). The results of these analyses (untabulated) do not show an association between CEO/CFO involvement or proprietary information and *READABILITY.*

[INSERT TABLE 5 HERE]

Comment Letter Response Tone

Table 6 reports the analysis results examining the association between misreporting firms and comment letter response tone. The regression results for each tone category are reported separately. I find a positive and statistically significant association between *NEGATIVE* and *AAER* ($p<0.05$). The tone analysis literature finds that a more negative tone in the 10-K is associated with poor financial performance (Davis and Tama-Sweet 2012) and fraud (Goel and Gangolly 2012; Lopatta et al. 2014). My findings related to comment letter responses by misreporting firms is consistent with these prior findings. I also find a moderately negative association ($p<0.10$) between *MODAL_STRONG* and *AAER*, suggesting that misreporting firms use less persuasive language or use fewer words that portray confidence or certainty in their comment letter responses than non-misreporting firms.

Supplemental Tone Analysis: As noted previously regarding SEC comment letter readability, SEC comment letter tone may also significantly influence the comment letter response tone. I include the SEC comment letter tone in the regression model with the corresponding comment letter response tone. I find (untabulated) a significant and positive association ($p<0.01$) between each SEC comment letter tone and the corresponding comment letter response tone. I also continue to find (untabulated) a positive and significant association

between *NEGATIVE* and *AAER* ($p < 0.10$) after controlling for negative tone in the SEC comment letter. I also examine whether the existence of proprietary information influences comment letter response tone, but do not find (untabulated) an association with any of tone variables.

[INSERT TABLE 6 HERE]

Comment Letter Process Rounds

Building upon the findings that misreporting firms provide comparatively more readable comment letter responses, I examine whether these responses, while clearer, adequately achieve the goal of avoiding additional scrutiny from the regulator during the comment letter review process. If a DCF reviewer is not satisfied with the initial comment letter response, additional comment letters and firm responses related to these same filings is issued are necessary, resulting in additional rounds of correspondence in the comment letter process. I use comment letter rounds (*ROUNDS*) as a proxy for the comment letter response adequacy. If misreporting firms provide comparatively more readable but inadequate responses, I expect these misreporting firms to endure more rounds in the comment letter process than non-misreporting firms.

To examine whether misreporting firm comment letter responses lead to comparatively more rounds in the comment letter process, I estimate model [1], substituting *ROUNDS* as the dependent variable in the model. It is important to note that this model controls for *NUM_TOPICS* and *NUM_FILINGS,* as a higher number of topics and filings addressed in the initial SEC comment letter will require a more comprehensive response and will likely lead to more rounds to sufficiently address all the issues raised. Table 7 presents the results of this analysis. Column (1) documents that *AAER* is positive and statistically significant ($p<0.05$), indicating that misreporting firms endure more rounds in the comment letter process than non-

misreporting firms. Column (2) presents this same analysis, while controlling for *READABILITY*. *AAER* continues to be positive and significant ($p<0.05$), while *READABILITY* is not significantly associated with *ROUNDS*. Taken together, these findings are consistent with misreporting firms providing inadequate responses to the initial SEC comment letter, requiring comparatively more rounds in the comment letter process. The readability of the comment letter response does not appear to be driving the need for more rounds for the sample firms.

[INSERT TABLE 7 HERE]

Misreporting Firm Comment Letter Responses During Non-Misreporting Periods

H1 predicts and finds that differences exist in the writing style of misreporting firms compared to non-misreporting firms when responding to SEC comment letters, specifically in the response readability and tone. To support these findings, I compare the comment letter response writing style for the sample misreporting firms during the misreporting periods to comment letter responses for these same firms during non-misreporting periods. Table 8 reports the mean differences between firms during the misreporting periods (*AAER*=1) and the non-misreporting periods (*AAER*=0). Similar to the results reported in Table 2, Panel C, I find that *NUM_FILINGS*, *NUM_TOPICS*, and *SEC_WORD_COUNT* are larger during the misreporting periods than non-misreporting periods ($p<0.01$). These results are consistent with the SEC finding more issues with the misreporting firms' public filings during the misreporting periods compared to the non-misreporting periods.

Consistent with the findings in Table 5, I find that the comment letter response *READABILITY* for the sample of misreporting firms is higher during the misreporting periods compared to the non-misreporting periods ($p<0.05$). I further find that firms use *MODAL_STRONG* words more frequently in comment letter responses during misreporting

periods than during non-misreporting periods. I do not find a significant difference in the propensity to use a *NEGATIVE* tone in comment letter responses during misreporting periods than non-misreporting periods. I further find that the *ROUNDS* is larger during the misreporting periods compared to non-misreporting periods ($p<0.01$). Overall, the results in Table 8 are consistent with firms using a different writing style to in their comment letter responses during misreporting periods compared to non-misreporting periods. Specifically, this writing style appears to be comparatively clearer, but the responses are less adequate leading to more rounds in the comment letter process.

[INSERT TABLE 8 HERE]

VI. CONCLUSION

I examine the association between misreporting firms and comment letter responses. More specifically, I explore whether misreporting firms have a different writing style, measured as readability and tone, when responding to SEC comment letters compared to non-misreporting firms.

I find that misreporting firms provide comment letter responses that are more readable than a propensity-score matched sample of non-misreporting firms, after controlling for factors related to firm characteristics, firm performance, and comment letter complexity. This finding is contrary to prior findings examining the readability of public filings by misreporting firms. Misreporting firms also provide comment letter responses that use a more negative tone than non-misreporting firms. While misreporting firms provide comparatively more readable comment letter responses, these responses appear to be less adequate and require more rounds in the comment letter process than non-misreporting firms. I further compare the writing style of

comment letter responses for the sample of misreporting firms during the misreporting periods to the comment letter responses for these same firms during non-misreporting periods. I continue to find that these firms provide comment letter responses that are more readable during misreporting periods than non-misreporting periods. I also find tone differences in the comment letter responses during misreporting periods compared to non-misreporting periods, supporting the prior findings that firms have a different comment letter response writing style during misreporting periods.

I use SEC AAERs to identify misreporting firms because this method is common in the fraud and misstatement literature (Beasley 1996; Carcello and Nagy 2004; Erickson, Hanlon, and Maydew 2006; Lennox and Pittman 2010; Dechow et al. 2011) and it identifies the most egregious misreporting violators. However, using SEC AAERs to identify misstating firms introduces certain limitations into the research design. Primarily, the use of AAER firms only identifies misstating events that the SEC chooses to investigate. Firms that are intentionally misstating their SEC filings, but are not identified in an AAER, may be included in the control sample of non-AAER firms. Further, the AAER sample may be biased due to the selection strategy the SEC uses in deciding which firms to investigate (Beasley et al. 2010; Dechow et al. 2011). For example, the SEC may only release information in an AAER for investigations where it is confident that fraud or other misstatements are present. As such, the use of AAERs to identify misstating firms reduces the instances of Type I error to a low level, but the potential for not identifying misstating firms (Type II error) is present and could introduce noise into the analysis that would bias against finding significant results.

I contribute to the SEC comment letter literature by identifying factors that contribute to firms adjusting the writing style of their comment letter responses. I also contribute to the fraud

textual analysis literature by examining the potential linguistic clues that misreporting firms provide in their comment letter responses. Prior fraud textual analysis research has focused on public filing primarily addressed to the investing public. This research seeks to understand whether the writing style of misstating firms reflects the same incentives when the target audience of their communication is the SEC instead of financial statement users.

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APPENDIX A
Variable Definitions

Variable Name	Variable Definition
AAER	Indicator variable equal to one if the firm received a comment letter related to a 10-K or 10-Q that is the subject of an SEC Accounting and Auditing Enforcement Release (AAER), zero otherwise.
AGE	The natural log of the number of years, through the year of the comment letter (year t), the firm has assets listed in Compustat.
AUDITOR_TENURE	The natural log of the number of consecutive years the firm has been audited by the same auditor.
BANKRUPTCY_RANK	The decile rank of the firm's Altman Z-score (Altman 1968).
BIG4	Indicator variable equal to one if the firm is audited by one of the Big 4 auditing firms (KPMG, PWC, Deloitte, E&Y), zero otherwise.
BMV	Book to market value, calculated as book value of equity / market value of equity.
DEBT_EQ_ISSUE	Indicitor variable equal to one if the sum of debt and equity issued exceeds two percent of total assets following the methodology in Lennox and Pittman (2010), zero otherwise.
DOMESTIC	Indicator variable equal to one if the firm is headquartered in the United States, zero otherwise.
EXTFINANCING	The firm's need for external financing, measured as the sum of equity financing and debt financing at time t+1, following Ettredge et al. (2011).
LOSS	Indicator variable equal to one if the firm reported income before extraordinary items less than zero, zero otherwise.
MULTIPLE_CL	Indicator variable equal to one if the firm has multiple comment letter response observations in the sample, zero otherwise
M&A	Indicator variable equal to one if the firm had an acquisition during the current year, zero otherwise.
NEG_BOOK_EQ	Indicator variable equal to one if the firm's total liabilities were greater than total assets, zero otherwise.
NUM_FILINGS	The number of firm filings (e.g. 10-K, 10-Q, 8-K) listed in the initial SEC Comment Letter.
NUM_TOPICS	The number of topics addressed in the initial SEC Comment Letter.
READABILITY	The readability of the firm's first comment letter response calculated as negative one multiplied by the Gunning Fog Index (Gunning 1952). This is done so that higher numbers represent more readable text.

RESTATE	Indicator variable equal to one if the firm filed a restatement in the two years prior to the receipt of the comment letter, zero otherwise
ROA	Return on assets, calculated as income before extraordinary items / total assets.
ROUNDS	The number of rounds in the comment letter process, measured as the number of SEC comment letters (FORM_FKEY = UPLOAD) in the comment letter conversation.
R&D	The proportion of resources devoted to R&D, calculated as R&D Expense / Total Assets.
SEC_WORD_COUNT	The natural log of the total number of words in the initial SEC comment letter.
SIZE	The natural log of total assets.
TONE	The firm's first comment letter response tone calculated as (number of tone words/total number of words) * 100. Measured using the following category word lists from Loughran and McDonald (2011): negative, positive, modal strong, and modal weak. See Appendix C for example words in each tone category.

APPENDIX B
Comment Letter Readability Examples

High Readability Comment Letter Response

Constellation Brands, Inc. April 2, 2007 Comment Letter Response. Fog Index = 15.252

> "Selling, general and administrative expenses increased to $612.4 million for Fiscal 2006 from $555.7 million for Fiscal 2005, an increase of $56.7 million, or 10%. The Constellation Wines segment's selling, general and administrative expenses increased $67.2 million primarily due to increased selling expenses of $31.6 million, general and administrative expenses of $26.8 million, and advertising expenses of $8.8 million to support the growth in the segment's business, primarily due to the costs related to the brands acquired in the Robert Mondavi acquisition. Included in the increase in general and administrative expenses was an expense of $6.4 million associated with the segment's U.K. defined benefit pension plan related to a reduction in the period over which unrecognized actuarial losses are amortized. The Constellation Beers and Spirits segment's selling, general and administrative expenses increased $4.6 million as increased selling expenses of $4.9 million and general and administrative expenses of $2.7 million were partially offset by lower advertising expenses of $3.0 million. The Corporate Operations and Other segment's selling, general and administrative expenses increased $7.0 million primarily due to costs associated with professional service fees incurred in connection with the Company's tender offer for Vincor that expired in December 2005 of $1.5 million and increased general and administrative expenses of $5.5 million to support the Company's growth."

Low Readability Comment Letter Response

Camelot Corp May 7, 2009 Comment Letter Response. Fog Index = 24.854

> "In determining the amount to accrue for the claim by Audio Visual Group dba AIMS Media, the Company considered the guidance in FASB 5 relating to lawsuits, which says ..."If an unfavorable outcome is determined to be reasonably possible but not probable, or if the amount of loss cannot be reasonably estimated, accrual would be inappropriate, but disclosure would be required...". Management's assessment of the judgment, which has been consistently disclosed in the same manner since 2001, was that it would be unlikely that the Company would ultimately pay out the entire amount of the judgment described in footnote 4. If there was to be a payout to this creditor, it would be at a reduced amount arrived at through negotiation with the creditor. At the time of the April 30, 2008 10-KSB filing, management had not yet commenced negotiations with the creditor, and therefore, the probable amount of the liability was not known nor was it accrued. Management believes that the provisions of FASB 5 were correctly applied in this matter. Management's disclosure has proven appropriate as subsequent to the period under review the Company settled this lawsuit for the sum of $40,000 only, which was fully disclosed on March 20, 2009 by an 8-K filing."

APPENDIX C
Example Tone Words

The following provides 20 examples of tone words included in each tone category taken from the keyword list created for Loughran and McDonald (2011). This list is available at http://www3.nd.edu/~mcdonald/Word_Lists.html.

Positive Tone

ACCOMPLISHMENTS	EFFECTIVE
ADVANCES	ENABLED
ASSURED	ENHANCEMENTS
ATTRACTIVENESS	ENTHUSIASM
BETTER	EXCITEMENT
BRILLIANT	FAVORABLY
COLLABORATORS	GOOD
CONSTRUCTIVE	HIGHEST
DELIGHTFULLY	IMPRESSING
DILIGENTLY	INCREDIBLY

Negative Tone

ANNOY	HARMS
CATASTROPHE	INADVERTENT
CONFINING	INSUFFICIENCY
CRITICISM	MALFEASANCE
DEGRADATION	MISTAKE
DETRIMENTAL	OPPORTUNISTIC
DISCOURAGED	PERVASIVE
DISSENTING	RATIONALIZATIONS
ENCROACHMENT	RETALIATED
FAILURES	STRAIN

Strong Modal Tone

ALWAYS	UNAMBIGUOUSLY
BEST	UNCOMPROMISING
CLEARLY	UNDISPUTED
DEFINITELY	UNDOUBTEDLY
DEFINITIVELY	UNEQUIVOCAL
HIGHEST	UNEQUIVOCALLY
LOWEST	UNPARALLELED
MUST	UNSURPASSED
NEVER	WILL
STRONGLY	

Weak Modal Tone

ALMOST	OCCASIONALLY
APPARENTLY	PERHAPS
APPEARS	POSSIBLE
CONCEIVABLE	SELDOM
COULD	SELDOMLY
DEPENDS	SOMETIMES
MAY	SOMEWHAT
MAYBE	SUGGEST
MIGHT	SUGGESTS
NEARLY	UNCERTAIN

APPENDIX D
Research Design Procedures

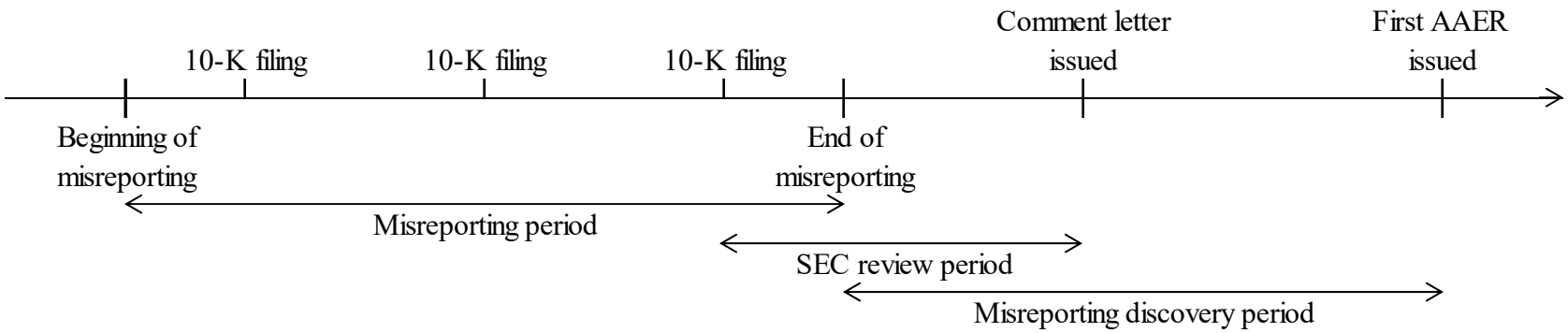
Procedures to Identify Incomplete Comment Letter Responses

Utilizing the AA comment letter database, I sort each comment letter conversation by file date. Following the methodology in Cassell et al. (2019), I eliminate comment letter conversations where (1) the file type of the first letter in the conversation is not from the SEC ("form_fkey" = "UPLOAD"), (2) the conversation is missing a "no further comment" letter from the SEC (i.e., "iss_wholet_keys" = 266), (3) the conversation only includes one letter from the SEC, (4) the conversation includes fewer than three letters, (5) the "file_date" of the first letter grouped by comment letter conversation ("cl_con_id") in the AA comment letter database does not match the "first_letter_date" of the corresponding "cl_con_id" in the AA comment letter conversations database indicating that the initial SEC letter is missing, (6) the "file_date" of the last letter grouped by comment letter conversation ("cl_con_id") in the AA comment letter database does not match the "last_letter_date" of the corresponding "cl_con_id" in the AA comment letter conversations database indicating that the last SEC comment letter is missing, and (7) the second letter in the conversation is not a response by the firm ("form_fkey"="CORRESP").

Procedures to Identity First Substantial Comment Letter Responses

Following the methodology in Cassell et al. (2019), I consider a comment letter response ("form_fkey" = "CORRESP") to not provide substantial information if it includes a request for extension ("iss_wholet_keys" = 253), the count of issues included in the letter ("list_cl_issue_taxgrp") is equal to zero or one and the letter indicates that it includes no new information ("iss_whole_let_keys" = 903), is a "tandy" letter ("iss_whole_let_keys" = 928), is a cover letter ("iss_whole_let_keys" = 1109), or is a duplicate letter ("iss_whole_let_keys" = 1257). I also consider a comment letter response to not be substantial if it contains less than 30 words.

FIGURE 1
Example timeline of misreporting, comment letter, and AAER issuance periods



Note: In this example, the firm engaged in a misreporting scheme impacting three 10-K filings. The last 10-K during the misreporting period was selected for SEC review and a comment letter was subsequently issued. In other instances, the first, second, or all three misstated 10-Ks could be selected for SEC review, potentially resulting in a comment letter issued for each 10-K filing reviewed. The initial comment letter is often issued within a few months of the 10-K filing (Cassell et al. 2013). In contrast, the discovery period and the subsequent decision by the SEC's Division of Enforcement to open an investigation often takes much longer, potentially enduring several months to several years after the last misstated 10-K was filed. Dechow et al. (2016) find that the median number of days between the fiscal year-end and the first comment letter is 193, or approximately 6.4 months. Conversely, Purda and Skillicorn (2015) find that the average length of time between when the misstatement occurred and the relevant AAER was issued is approximately 3.4 years.

TABLE 1
Sample Selection

Panel A: AAER-Related Comment Letters

Unique firms cited in AAERs issued between January 2005 and September 2014	**339**
Less: AAER firms with no available CIK number	(10)
Less: AAER is related to an issue other than 10-K or 10-Q misreporting	(6)
Less: AAER only implicates the auditor	(4)
AAER firms with a misreported 10-K or 10-Q	**319**
Less: AAER firms without any comment letter data in Audit Analytics	(66)
Less: AAER firm comment letter is not related to misstated 10-K or 10-Q	(164)
Add: AAER firms receiving multiple comment letters related to misstated AAER period	23
Less: observations without sufficient control variable data	(9)
Sample of comment letter conversations related to AAER periods	**103**

Panel B: Comment Letter Conversations Control Sample

Unique comment letter conversations per Audit Analytics as of June 2016	**95,926**
Less: conversations not related to 10-K filings	(66,555)
Less: incomplete conversations in Audit Analytics (see Appendix E)	(12,689)
Less: observations without sufficient control variable data	(2,580)
Less: AAER comment letter sample	(103)
Sample of non-AAER comment letter conversations	**13,999**
Less: observations for years not represented in AAER sample (see Table 2, Panel A)	(3,687)
Sample of non-AAER comment letter conversations used for matched-sample analyses	**10,312**

TABLE 2
Descriptive Statistics

Panel A: Firms by Year Addressed in SEC Comment Letter

	AAER Sample		Non-AAER Sample		Total	
Year	**Count**	**Percentage**	**Count**	**Percentage**	**Count**	**Percentage**
2003	1	1.0%	22	0.2%	23	0.2%
2004	27	26.2%	661	6.4%	688	6.6%
2005	21	20.4%	838	8.1%	859	8.3%
2006	15	14.6%	1,050	10.2%	1,065	10.2%
2007	16	15.5%	1,279	12.4%	1,295	12.4%
2008	10	9.8%	1,552	15.1%	1,562	15.0%
2009	9	8.7%	1,680	16.3%	1,689	16.2%
2010	2	1.9%	1,537	14.9%	1,539	14.8%
2011	2	1.9%	1,693	16.4%	1,695	16.3%
Total	103	100.0%	10,312	100.0%	10,415	100.0%

Panel B: Sample Frequency by Industry

	AAER Sample		Non-AAER Sample		Total	
Industry	**Count**	**Percentage**	**Count**	**Percentage**	**Count**	**Percentage**
Agriculture	0	0.0%	31	0.3%	31	0.3%
Mining and Construction	3	2.9%	377	3.7%	380	3.6%
Food	6	5.8%	189	1.8%	195	1.9%
Textiles and Printing/Publishing	3	2.9%	350	3.4%	353	3.4%
Chemicals	0	0.0%	242	2.3%	242	2.3%
Pharmaceuticals	5	4.9%	636	6.2%	641	6.2%
Extractive industries	1	1.0%	422	4.1%	423	4.1%
Durable Manufacturers	15	14.6%	1,867	18.1%	1,882	18.1%
Computers	21	20.4%	1,395	13.5%	1,416	13.6%
Transportation	3	2.9%	708	6.9%	711	6.8%
Utilities	4	3.8%	452	4.4%	456	4.4%
Retail	14	13.6%	812	7.9%	826	7.9%
Financial Institutions	14	13.6%	1,324	12.8%	1,338	12.8%
Insurance and Real Estate	4	3.9%	649	6.3%	653	6.3%
Services	10	9.7%	787	7.6%	797	7.6%
Other	0	0.0%	71	0.7%	71	0.7%
Total	103	100.0%	10,312	100.0%	10,415	100.0%

Panel C: Comparison of AAER Firm vs. Non-AAER Firm Observations

	(*AAER* = 1)						(*AAER* = 0)							
VARIABLES	**N**	**Mean**	**SD**	**25%**	**50%**	**75%**	**N**	**Mean**	**SD**	**25%**	**50%**	**75%**	**t-stat**	
READABILITY	103	-19.056	1.796	-20.205	-18.889	-17.832	10312	-19.453	1.794	-20.597	-19.400	-18.215	-2.237	**
NEGATIVE	103	1.481	0.963	0.807	1.318	1.846	10312	1.276	0.747	0.731	1.099	1.642	-2.768	***
POSITIVE	103	0.416	0.282	0.229	0.364	0.549	10312	0.397	0.290	0.197	0.354	0.549	-0.641	
MODAL_STRONG	103	0.369	0.245	0.219	0.334	0.515	10312	0.360	0.230	0.198	0.333	0.490	-0.390	
MODAL_WEAK	103	0.308	0.190	0.161	0.279	0.396	10312	0.325	0.211	0.179	0.278	0.420	0.788	
SIZE	103	6.549	2.637	4.618	6.683	8.688	10312	6.511	2.733	4.929	6.733	8.404	-0.139	
AGE	103	17.282	15.319	6.000	13.000	21.000	10312	19.880	15.049	9.000	15.000	26.000	1.743	*
NUM_FILINGS	103	1.883	0.963	1.000	2.000	3.000	10312	1.640	0.830	1.000	1.000	2.000	-2.963	***
NUM_TOPICS	103	11.350	7.387	6.000	10.000	17.000	10312	9.891	6.683	5.000	8.000	14.000	-2.202	**
SEC_WORD_COUNT	103	1,306.631	676.753	849.000	1,086.000	1,562.000	10312	1,124.293	560.304	745.000	977.000	1,347.000	-3.279	***
RESTATE	103	0.272	0.447	0.000	0.000	1.000	10312	0.148	0.355	0.000	0.000	0.000	-3.501	***
LOSS	103	0.233	0.425	0.000	0.000	0.000	10312	0.318	0.466	0.000	0.000	1.000	1.844	*
BANKRUPTCY_RANK	103	7.194	2.525	6.000	8.000	9.000	10167	5.907	2.644	4.000	6.000	8.000	-4.918	***
EXTFINANCING	103	-0.009	0.176	-0.069	0.000	0.029	10312	0.021	0.287	-0.056	-0.002	0.029	1.032	
BMV	103	0.456	0.820	0.241	0.377	0.656	10312	0.533	0.994	0.222	0.470	0.805	0.783	
ROA	103	0.002	0.243	0.006	0.037	0.086	10312	-0.146	0.975	-0.024	0.023	0.068	-1.542	
BIG4	103	0.660	0.476	0.000	1.000	1.000	10312	0.717	0.450	0.000	1.000	1.000	1.280	
ROUNDS	103	3.126	1.117	2.000	3.000	4.000	10312	2.734	1.031	2.000	2.000	3.000	-3.841	***

Note: See Appendix A for variable definitions. *AGE* and *SEC_WORD_COUNT* are presented in their raw (unlogged) form. T-statistics of mean differences are included at the right. Two-tailed p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.10.

TABLE 3
Correlations

	VARIABLES	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1)	*AAER*												
(2)	*READABILITY*	**0.025**											
(3)	*NEGATIVE*	**0.025**	**-0.029**										
(4)	*POSITIVE*	0.007	**0.059**	**0.040**									
(5)	*MODAL_STRONG*	0.007	**0.039**	**0.030**	**0.068**								
(6)	*MODAL_WEAK*	-0.006	**-0.177**	**0.216**	**-0.046**	**0.079**							
(7)	*SIZE*	0.006	-0.011	**0.058**	0.003	**-0.033**	**-0.020**						
(8)	*AGE*	**-0.018**	**-0.033**	**0.063**	**0.042**	**0.023**	**0.044**	**0.058**					
(9)	*NUM_FILINGS*	**0.037**	**0.158**	**0.017**	**0.086**	**0.113**	**-0.051**	-0.016	**-0.018**				
(10)	*NUM_TOPICS*	**0.036**	**0.366**	**0.049**	**0.104**	**0.193**	**-0.093**	0.002	-0.008	**0.339**			
(11)	*SEC_WORD_COUNT*	**0.046**	**0.385**	**0.092**	**0.112**	**0.137**	**-0.083**	**0.094**	**-0.025**	**0.345**	**0.734**		
(12)	*RESTATE*	**0.029**	**0.028**	**0.031**	0.012	-0.004	-0.015	-0.013	**-0.040**	**0.033**	**0.059**	**0.056**	
(13)	*LOSS*	-0.015	0.007	**0.038**	0.012	0.016	-0.011	**-0.054**	**-0.196**	**0.032**	**0.036**	**0.056**	**0.08**
(14)	*BANKRUPTCY_RANK*	**0.043**	-0.011	**-0.122**	-0.002	**0.032**	**0.031**	**-0.125**	**0.080**	-0.009	**0.032**	**-0.031**	**-0.052**
(15)	*EXTFINANCING*	-0.007	0.002	**-0.019**	0.000	0.004	0.005	-0.015	**-0.126**	0.007	0.005	**0.024**	**0.039**
(16)	*ROA*	0.013	-0.004	**0.024**	0.010	0.000	**-0.022**	**0.024**	**0.122**	-0.005	0.005	-0.007	**-0.037**
(17)	*BMV*	-0.007	**0.034**	**0.049**	0.000	0.002	-0.013	**0.041**	0.013	**0.020**	-0.002	**0.029**	**-0.023**
(18)	*BIG4*	**-0.017**	**-0.055**	**0.028**	**0.018**	**-0.048**	**0.034**	**0.080**	**0.211**	**-0.064**	**-0.059**	**-0.053**	**-0.033**
(19)	*ROUNDS*	**0.041**	**0.161**	0.014	**0.058**	-0.016	**-0.046**	**0.034**	**-0.033**	**0.096**	**0.268**	**0.332**	0.012

TABLE 3
Correlations (continued)

	VARIABLES	(13)	(14)	(15)	(16)	(17)	(18)
(14)	*BANKRUPTCY_RANK*	**-0.333**					
(15)	*EXTFINANCING*	**0.266**	**-0.211**				
(16)	*ROA*	**-0.314**	**0.280**	**-0.446**			
(17)	*BMV*	**-0.040**	0.011	**-0.087**	**0.221**		
(18)	*BIG4*	**-0.218**	**0.125**	**-0.214**	**0.243**	**0.027**	
(19)	*ROUNDS*	**0.049**	**-0.033**	**0.039**	**-0.046**	0.001	**-0.086**

Notes: This table presents the Pearson correlations. Variable definitions are provided in Appendix A. Bolded correlations are significant at $p < 0.05$.

TABLE 4
Propensity-Score Matching

Panel A: Propensity-Score Match Logistic Regression

Variable	Coeff.	z-stat	p-value	
SIZE	0.202	3.382	0.001	***
AGE	-0.357	-1.961	0.050	**
LOSS	0.027	0.093	0.926	
BANKRUPTCY_RANK	0.275	5.027	0.000	***
BIG4	-0.780	-2.910	0.004	***
AUDITOR_TENURE	-0.021	-0.135	0.892	
M&A	0.768	2.886	0.004	***
NEG_BOOK_EQ	0.346	0.566	0.571	
DEB_EQ_ISSUE	0.260	1.198	0.231	
Intercept	-4.267	-3.591	0.000	***
N	10,234			
Pseudo R^2	0.163			
Year FE	Included			
Industry FE	Included			
Robust SE	Included			
Area Under ROC Curve	0.841			

Notes: This table presents the estimation results for model [2]. Variable definitions are provided in Appendix A. Reported standard errors are reported in parentheses. *, **, and *** indicate two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

Panel B: Covariate Balance

	(AAER = 1)		(AAER = 0)				
Variable	**N**	**Mean**	**N**	**Mean**	**Difference**	**t-statistic**	**p-value**
SIZE	103	6.549	103	6.486	-0.063	-0.171	0.865
AGE	103	2.473	103	2.532	0.059	0.489	0.625
LOSS	103	0.233	103	0.184	-0.049	-0.855	0.394
BANKRUPTCY_RANK	103	7.194	103	7.194	0.000	0.000	1.000
BIG4	103	0.660	103	0.621	-0.039	-0.579	0.564
AUDITOR_TENURE	103	7.680	103	7.233	-0.447	-0.464	0.643
M&A	103	0.165	103	0.184	0.019	0.365	0.715
NEG_BOOK_EQ	103	0.029	103	0.039	0.010	0.383	0.702
DEB_EQ_ISSUE	103	0.670	103	0.631	-0.039	-0.582	0.561

Note: See Appendix A for variable definitions. Two-tailed p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.10.

TABLE 5
The Association Between Misstating Firms and Comment Letter Response Readability

DV - *READABILITY*	
VARIABLES	
AAER	**0.661****
	(0.300)
SEC_READABILITY	
SIZE	0.034
	(0.086)
AGE	0.130
	(0.146)
NUM_FILINGS	0.104
	(0.132)
NUM_TOPICS	0.063***
	(0.022)
SEC_WORD_COUNT	1.021***
	(0.377)
RESTATE	-0.380
	(0.259)
LOSS	-0.520
	(0.363)
BANKRUPTCY_RANK	0.006
	(0.058)
EXTFINANCING	0.144
	(0.625)
BMV	0.207**
	(0.094)
ROA	-0.693
	(0.596)
BIG4	-0.372
	(0.371)
MULTIPLE_CL	-0.481
	(0.369)
Intercept	-27.605***
	(3.323)
Observations	206
Year FE	Included

Industry FE	Included
Adjusted R^2	0.235

Notes: This table presents the estimation results for model [1] with *READABILITY* as the dependent variable. Variable definitions are provided in Appendix A. Reported standard errors are reported in parentheses. *, **, and *** indicate two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

TABLE 6
The Association Between Misstating Firms and Comment Letter Response Tone

DV =	*NEGATIVE*	*POSITIVE*	*MODAL_STRONG*	*MODAL_WEAK*
VARIABLES	**(1)**	**(2)**	**(3)**	**(4)**
AAER	**0.292****	**0.020**	**-0.065***	**0.014**
	(0.141)	**(0.055)**	**(0.036)**	**(0.036)**
SIZE	0.003	-0.004	-0.013	0.003
	(0.037)	(0.014)	(0.012)	(0.010)
AGE	0.215**	-0.013	-0.005	0.014
	(0.084)	(0.029)	(0.024)	(0.018)
NUM_FILINGS	-0.048	-0.013	0.021	-0.003
	(0.053)	(0.022)	(0.020)	(0.016)
NUM_TOPICS	0.019	0.002	-0.000	-0.003
	(0.015)	(0.004)	(0.003)	(0.003)
SEC_WORD_COUNT	-0.298	0.008	0.021	0.002
	(0.331)	(0.081)	(0.059)	(0.043)
RESTATE	0.040	0.021	0.090**	-0.005
	(0.114)	(0.050)	(0.038)	(0.036)
LOSS	0.307*	0.098	-0.015	-0.001
	(0.177)	(0.078)	(0.060)	(0.038)
BANKRUPTCY_RANK	-0.032	0.010	-0.006	-0.001
	(0.031)	(0.014)	(0.010)	(0.006)
EXTFINANCING	-0.503	-0.039	-0.063	0.103
	(0.321)	(0.143)	(0.093)	(0.085)
BMV	0.020	-0.005	0.004	0.022*
	(0.044)	(0.035)	(0.022)	(0.013)
ROA	0.366	0.049	-0.062	-0.096
	(0.238)	(0.154)	(0.076)	(0.075)
BIG4	0.205	0.088	0.034	0.039
	(0.174)	(0.072)	(0.054)	(0.035)
MULTIPLE_CL	0.093	-0.109	0.089	0.031
	(0.222)	(0.067)	(0.057)	(0.047)
Intercept	3.166	0.489	0.328	0.015
	(2.325)	(0.593)	(0.431)	(0.290)
Observations	206	206	206	206
Year FE	Included	Included	Included	Included
Industry FE	Included	Included	Included	Included

Adjusted R^2	0.184	0.061	0.021	0.036

Notes: This table presents the estimation results for model 1] with the tone variables *NEGATIVE*, *POSITIVE*, *MODAL_STRONG*, and *MODAL_WEAK* as the dependent variables in columns (1), (2), (3), and (4), respectively. Variable definitions are provided in Appendix A. Reported standard errors are reported in parentheses. *, **, and *** indicate two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

TABLE 7
The Association Between Misstating Firms and Total Rounds in Comment Letter Process

DV = *ROUNDS*		
VARIABLES	(1)	(2)
AAER	**0.443****	**0.446****
	(0.175)	**(0.179)**
READABILITY		-0.005
		(0.042)
SIZE	0.064	0.064
	(0.040)	(0.041)
AGE	-0.036	-0.035
	(0.095)	(0.096)
	-0.039	-0.039
	(0.076)	(0.076)
NUM_TOPICS	-0.003	-0.003
	(0.020)	(0.020)
SEC_WORD_COUNT	1.101***	1.107***
	(0.339)	(0.340)
RESTATE	-0.175	-0.177
	(0.153)	(0.154)
LOSS	0.293	0.290
	(0.221)	(0.221)
BANKRUPTCY_RANK	0.042	0.042
	(0.036)	(0.036)
EXTFINANCING	-0.791**	-0.790**
	(0.375)	(0.377)
BMV	-0.126**	-0.125**
	(0.062)	(0.062)
ROA	0.517*	0.513*
	(0.304)	(0.305)
BIG4	-0.413**	-0.415**
	(0.202)	(0.205)
MULTIPLE_CL	-0.185	-0.188
	(0.209)	(0.209)
Intercept	-6.515***	-6.659**
	(2.344)	(2.588)
Observations	206	206

Year FE	Yes	Yes
Industry FE	Yes	Yes
Adjusted R2	0.209	0.204

Notes: This table presents the estimation results for model [1] with *ROUNDS* as the dependent variable. Column (2) includes *READABILITY* as an additional control variable. Variable definitions are provided in Appendix A. Reported standard errors are reported in parentheses. *, **, and *** indicate two-tailed statistical significance at the 10%, 5%, and 1% levels, respectively.

TABLE 8
Analysis of Mean Differences of AAER Firms during Misreporting Periods Compared to Non-misreporting Periods

	Misreporting Period		Non-misreporting Period					
VARIABLES	**N**	**Mean**	**N**	**Mean**	**Difference**	**t-statistic**	**p-value**	
READABILITY	103	-19.056	381	-19.456	-0.401	-2.092	0.037	**
NEGATIVE	103	1.481	381	1.404	-0.078	-0.874	0.383	
POSITIVE	103	0.416	381	0.401	-0.014	-0.433	0.665	
MODAL_STRONG	103	0.369	381	0.311	-0.057	-2.433	0.015	**
MODAL_WEAK	103	0.308	381	0.316	0.008	0.381	0.704	
SIZE	103	6.549	381	8.250	1.701	5.967	0.000	***
AGE	103	17.282	381	28.554	11.272	6.016	0.000	***
NUM_FILINGS	103	1.883	381	1.572	-0.311	-3.368	0.001	***
NUM_TOPICS	103	11.350	381	8.945	-2.405	-3.279	0.001	***
RESTATE	103	0.272	381	0.244	-0.028	-0.576	0.565	
LOSS	103	0.233	381	0.262	0.029	0.607	0.544	
BANKRUPTCY_RANK	103	7.194	380	5.542	-1.652	-5.991	0.000	***
EXTFINANCING	103	-0.009	381	-0.002	0.006	0.252	0.801	
BMV	103	0.456	381	0.555	0.099	0.903	0.367	
ROA	103	0.002	381	-0.091	-0.093	-1.032	0.302	
BIG4	103	0.660	381	0.885	0.224	5.631	0.000	***
ROUNDS	103	3.126	381	2.693	-0.433	-3.860	0.000	***

Note: See Appendix A for variable definitions. Two-tailed p-values are denoted as follows: *** p<0.01, ** p<0.05, * p<0.10.